

Bike Sharing: Forecasts to Decisions

Decision brief

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Decision in one paragraph

The study supports a transparent discussion of forecast error and planning preferences. It does **not support operational deployment on its own**. The selected calendar model has useful predictive structure, but its uncertainty bands under-cover the final quarter, especially at high predicted demand. A cost-aware target illustrates the trade-off between planning too little and too much. Its costs are assumptions, and the data do not identify actual savings.

What was evaluated

The audited dataset contains aggregate hourly rentals in 2011–2012. Four interpretable candidates were compared in July–September 2012. Forecasts were issued weekly using only preceding observations. Poisson calendar was selected by mean absolute error (MAE); signed residual adjustments supplied the planning policy. October–December was reserved until all choices were frozen.

The final assessment contains 2,168 observed hours and 40 absent hours. Missing hours were excluded, not filled with zero. Earlier test observations could enter later scheduled fits and calibration windows; no test outcome selected a model, policy, cost ratio or method setting. This is sequential evaluation of a fixed procedure, not one forecast frozen for a whole quarter.

Point error and uncertainty

Period	Hours	MAE	RMSE
Validation	2,208	53.268	83.215
Final test	2,168	61.378	95.811

MAE is the average absolute discrepancy in rentals per hour. Final-test error is higher than validation error. Only the previously selected model was tested, so this is not a new competition among all candidates.

Nominal	Validation	Final test	Test width
80%	81.25%	75.69%	150.273
90%	90.44%	85.19%	207.771
95%	94.75%	91.42%	267.424

Coverage is the fraction of observations within the nominal prediction band; width is measured in rental counts. The nominal 90% band covers only 85.19% overall and 74.81% in the high predicted-demand group. Recent forecast errors are therefore not a sufficiently reliable uncertainty model across these conditions. No method was retuned after seeing this result. The procedure has no universal or conditional coverage guarantee.

Planning trade-off

Choose a nonnegative continuous hourly workload target, measured in rental-equivalent units. Underprediction incurs an assumed cost ratio r ; overprediction costs one normalized unit. Loss is $r \max(y - a, 0) + \max(a - y, 0)$, for observed rentals y and target a . These are proxy costs, not euros. Fractional targets are allowed because the quantity is a workload allowance, not a count of bikes or staff.

The validation-selected policy adjusts Poisson forecasts using the $r/(1+r)$ quantile of past signed standardized errors. It was chosen from five policies in every scenario. Only those saved choices were tested.

Cost ratio	Validation cost	Test cost	Test target
0.25	25.385	33.017	185.639
0.5	36.897	45.849	205.390
1	52.328	62.766	229.198
2	71.368	81.273	253.693
4	90.937	102.194	275.921
9	115.067	127.825	301.594

Compare periods cautiously: season and observations differ. Compare policies within one cost scenario, not losses across different cost ratios. Higher penalties for underprediction produce larger targets and more excess planning. At ratio 9, targets fall below final-test rentals in 10.10% of hours; that is an observed frequency, not a guaranteed service level.

At equal costs the adjusted policy has final loss 62.766, while the separately frozen Poisson point forecast has MAE 61.378. Thus the small validation advantage does not persist there. We retain the selected policy and report the adverse finding rather than select again on the test quarter.

What would be needed for an operational decision?

Obtain defensible marginal costs and a mapping from rental-equivalent workload to actual resources. Station availability, unmet requests, trip movements and capacity constraints are needed for rebalancing or lost-demand analysis. A design that measures the effect of changing operations is needed for savings claims. Observed rentals alone cannot provide that evidence.

The historical notebook explored the full dataset before this modernization. The maintained code preserves temporal separation, but cannot establish that no human ever inspected these outcomes. Data lineage, physical weather units and timezone qualifications remain documented. One two-year dataset is not evidence of robustness across locations or future operating regimes.

Evidence and reproducibility

github.com/DiogoRibeiro7/bike-sharing contains the source, audited provenance, frozen selections, final reports and exact commands. Tables are generated from committed JSON evidence, not copied from a spreadsheet. With Python 3.12 and the locked Poetry environment, run `poetry run python scripts/reproduce_release.py` to verify the frozen assessment. Replays check reproducibility; they do not create another untouched test or justify tuning to the reported results.